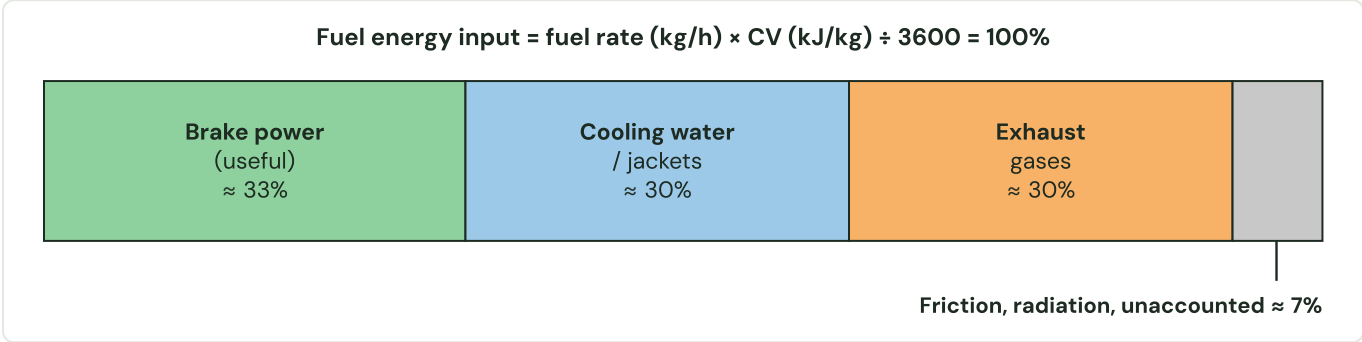
UNIT 4 · TOPIC 4(A)

Heat Load and Heat Balance

Syllabus topic: Heat load and power transmission from piston to the flywheel



Heat balance of a farm diesel at rated load (indicative; redrawn from the source figure)

$$\text{Fuel energy input (kW)} = \text{fuel rate (kg/h)} \times \text{CV (kJ/kg)} \div 3600$$

Share of fuel energy at rated load (indicative)	Value
Brake power	~30–35%
Cooling system	~30%
Exhaust	~30%
Friction, radiation, unaccounted	~5–10%

Rule of thumb: heat rejected to the coolant ≈ engine brake power — the basis of radiator sizing (heat LOAD).

Cooling load

$$Q\ (\text{kW}) = \dot{m}_w \times c_p \times \Delta T$$

Example. Coolant flow 60 L/min = 1 kg/s, c_p = 4.187 kJ/kg·K, inlet–outlet ΔT = 7 °C → Q = 1 × 4.187 × 7 = **29.3 kW** (matches a ~30-kW engine).

On the air side, the radiator + fan must reject the same heat to the air (LMTD/effectiveness — Unit 9 instrumentation).

Heat balance sheet — worked example

Fuel 10 kg/h, CV 42,500 kJ/kg → input = 10 × 42,500/3600 = **118 kW**.

Item	kW	Share
Brake power	40	33.9%
Cooling water	35	29.7%
Exhaust	34	28.8%
Radiation / unaccounted (balance)	9	7.6%
Total	118	100% — the sheet must close

Practice MCQs

Q91. At rated load, the useful (brake) share of fuel energy is roughly:

(a) 10% (b) 33%
(c) 60% (d) 90%

Q92. Heat carried away by cooling water is roughly:

(a) 5% (b) 15%
(c) 30% (d) 60% of fuel energy

Q93. Fuel 10 kg/h of CV 42,500 kJ/kg → energy input ≈

(a) 68 kW (b) 88 kW
(c) 118 kW (d) 425 kW

Q94. Coolant flow 1 kg/s with ΔT = 7 °C removes about:

(a) 7 kW (b) 15 kW
(c) 29 kW (d) 58 kW

Q95. Radiator heat load is roughly equal to:

(a) Twice IP (b) The engine's brake power
(c) Exhaust heat only (d) Zero

Answers

Q91 — (b) 33%
Q92 — (c) 30%
Q93 — (c) 118 kW — *10 × 42,500/3600 = 118 kW.*
Q94 — (c) 29 kW — *1 × 4.187 × 7 = 29.3 kW.*
Q95 — (b) The engine's brake power

Sources: Liljedahl et al. — Tractors and their Power Units; standard IC-engine texts; Kepner et al.; NPTEL farm power lectures.